Infosys — Aptitude Test 1

Questions 1–25 of 100 · 25 questions

Question 1. Find the missing number in the series. 2, 5, 10, 17, ? , 41.

Solution:
The difference between the successive
terms are a sequence of prime numbers.
2 ~ 5 = 3;
5 ~ 10 = 5;
10 ~ 17 = 7
17 ~ ? = 11;
? ~ 41 = 13
Missing number is 28

Question 2. Find the missing number in the series. 8 : 18 : : 24 : ?

Solution:
8 = 32 – 1;
18 = 42 + 2
24 = 52 – 1;
? = 62 + 2
Answer = 38

Question 3. Find the missing number in the series. 7,14,55,110,?

Solution:
7 + reverse of 7 = 7 + 7 = 14
14 + reverse of 14 = 14 + 41 = 55
55 + reverse of 55 = 55 + 55 = 110
110 + reverse of 110
= 110 + 011 = 121

Question 4. Find the missing number in the series. 2,4,7,10,15,18,....

Solution:
0 + 2 = 2;
1 + 3 = 4;
2 + 5 = 7
3 + 7 = 10;
4 + 11 = 15;
5 + 18= 18
6 + 17 = 33
Adding prime numbers to (0,1,2,3, …)

4.5

0.5
1.5

only one way.
The three women can be arranged in 3!
ways.

Total number of arrangements:
= 1 x 3! = 6

Question 5. Three members went to a shop and took 3 kerchiefs each costs 10 Rs. Total price is Rs 30. After they went off the owner of the sop realizes that he know the 3 members, so he gives the worker 5 Rs and tells to return them. The worker gives them Rs 5, then they gives 2 Rs tip to him and remaining 3 Rs they share Rs 1. Expense of each person is Rs. 9. Total expense of three persons is Rs. 27. Tips = Rs. 2. Overall total = Rs. 29. Where is the remaining 1 Rs?

Solution:
Total Expense Rs. 27 includes the tip.
One should not add Rs. 2 with Rs. 27.
Expense for kerchief = Rs. 25
Tip = Rs. 2
Balance = Rs. 3
Hence tallied.

I n f o s y s

Question 6. A man has two ropes of varying thickness (Those two ropes are not identical, they aren't the same density nor the same length nor the same width). Each rope burns in 30 minutes. He actually wants to measure 45 minutes. How can he measure 45 minutes using only these two ropes. He can't cut the one rope in half because the ropes are non-homogeneous and he can't be sure how long it will burn.

Solution:
Take the first rope and light it in both ends. It will burn out in 15 minutes.
Take the second rope and light it in one

end at the same moment when the first rope burns out.

Question 7. There are 6561 balls out of them 1 is heavy. Find the minimum number of times the balls have to be weighed for finding out the heavy ball.

Solution:
Split 6561 into 3 groups of 2687.
From this take two groups and place them on the two plates of the weight balance. If the heavy ball is on one of the plates, take that group and do the above process again.
If both plates are equal, take the group which was kept aside and do the same process.
Number of weighing = 8.

Shortcut:
If number of balls
= 3n ,
number of weighing = n
6561 = 38 .

Question 8. A mango vendor travels between two cities through 30 toll booths. He had 90 mangoes in 3 baskets, each basket with a maximum capacity of 30 mangoes. He has to pay the toll of 1 mango per basket per booth. If the vendor is intelligent, how many mangoes will he have when he reaches the destination?

tractors as they have. After some days C gives A and B as many tractors as they have. Finally each has 24 tractors. What is the original Number of tractors each had in the beginning?

I n f o s y s

Solution:
From the final statement, total number of tractors = 3 x 24 = 72.
Also
from
the
final
statement,
the
number of tractors of A and B got doubled.
Number of tractors with A, B and C, before C lends them is:
A = 12; B = 12; C = 48

From second statement, when B lends to A and C, the number of tractors with A and C gets doubled.
Number of tractors with A, B and C,
before B lends them is:
A = 6; B = 42; C = 24
From first statement, when A lends tractors to B and C, the number of tractors with B and C gets doubled.
Number of tractors with A, B and C,
before A lends them is:
A = 39; B = 21; C = 12

Solution:
After 10 toll booths he would have given 30 mangoes. So, one basket will be empty. He must throw that basket away.
After another 15 booths, he must throw the second basket.
Now he has 30 mangoes left in one basket and 5 booths left.
He will have 25 mangoes left at the end of the journey.

8.33

200
400
St
Sc
+

St : Sc = 3 : 4

Question 9. GOOD is coded as 164, BAD is coded as 21. If UGLY coded as 260 then JUMP?

Solution:
Take the position of each letter in the English alphabet.
G = 7; O = 15;
D = 4
G+O+O+D = 7 + 15 + 15 + 4 = 41
Sum x Number of letters in the word,
41 x 4 = 164
B+A+D = 3(2 + 1 + 4) = 21
U+G+L+Y = 4(21 + 7 + 12 + 25)
= 260
Similarly,
J+U+M+P = 4(10 + 21 + 13 + 16)
= 240

Question 10. If HAT = 58, then KEEP = ?

Question 11. There are 3 societies A, B, C having some tractors each. A Gives B and C as

many tractors as they already have. After some days B gives A and C as many

= 100y + x
100y + x – 5 = 2(100x + y)
100y + x – 200x – 2y = 5
98y = 199x + 5
98y = 196x + (3x + 5)
y = 2x + (3x + 5)/98
(3x + 5)/98 = integer
If (3x+ 5)/98 = 1, x = 31
y = 63.
The Cheque was given for 31 Rupees and 63 paise.

Question 12. Find the greatest number that will divide 45, 91 and 183 so as to leave the same remainder in each case.
Solution:
Answer will be the HCF of the differences
HCF [(91-45), (183-91)] = HCF(46, 92) = 46

Question 13. A person goes to a bank and quotes x rupees and y paise on a Cheque. The cashier misreads it and gives y Rs and x paise. The man comes out and donates 5 paise to a beggar. Now, the man has exactly double the amount he has quoted on the Cheque.

Solution:
H = 8; A = 1; T = 20
H+A+T = 29
HAT = 2 x 29 = 58
K+E+E+P = 11 + 5 + 5 + 16 = 37
KEEP = 2 x 37 = 74

Solution:
The actual amount quoted in terms of paise
= 100x + y
The amount yielded by the banker

Question 14. If abcde is a five digit number and abcde*4=edcba, then what is the value of b+c+d? (number of each letter is unique).

Solution:
Value of ‘a’ cannot be more than 2.
Otherwise the product will end in a 6 digit number.

a ≠0, if so, the number will be a four digit.
a ≠1, because, any number multiplied with an even number will give even number in the unit digit. (unit digit of “edcba” is ‘a’).
So, a = 2.
e x 4 = 8.
e = 3 or 8.
e ≥8, because a = 2; so e = 8
There is no carry over when b is multiplied with 4, because 2 x 4 = 8, where 2 = a
and 8 = e.
So b = 0 or 1. b ≠2 because a = 2.
e x 4 = 8 x 4 = 32.
3 is carry over.
4d + 3 = b
Unit digit of 4d = 7 or 8 to get b = 0 or 1.
Unit digit of 4d = 8 because 4 is even number
d = 2 or 7 so that 4 x 2 = 8 or 4 x 7 = 28
d ≠2 because, a = 2; so d = 7
4d + 3 = 4 x 7 + 3 = 28+3 = 31.
So, b = 1
3 is carry over.
4c + 3 = c and must have 3 as carry over, because 4b + 3 = 7 = d
The only value of c satisfying the above is 9.

4 x c + 3 = 4 x 9 + 3 = 39.
a = 2; b= 1; c = 9; d = 7; e = 8
b + c + d = 1 + 9 + 7 = 17

I n f o s y s

Question 15. Lucia is a wonderful grandmother her age is between 50&70. Each of her sons have as many sons as they have brothers. Their combined number gives Lucia’s present age. What is the age of Lucia?

Solution:
Assume the number of sons for Lucia = n.
Number of brothers for each son = n-1.
Number of sons for each son of Lucia = n-1
Total grandsons of Lucia = n(n-1)
= n2 – n
Combined number of sons and

grandsons
= n + n2 – n
= n2

n is number of sons, so it must be an integer.
A perfect square between 50 and 70 is 64.
Age of Lucia = 64.
Number of sons = 8; grandsons = 56

Question 16. In a soap company a soap is manufactured with 11 parts. For making one soap you will get 1 part as scrap. At the end of the day you have 251 such scraps. From that how many soaps can be manufactured?

Solution:
Effective number of parts per soap = 10
Number of soaps = 251/10 = 25 soaps
One small part will be remaining.

Note:
If total number of parts(n) is not a multiple of number of effective parts(e), then
Number of products = quotient of (n/e)
If total number of parts(n) is a multiple of number of effective parts(e), then
Number of products = (n/e) – 1.

Question 17. There is well of depth 30m and frog is at bottom of the well. He jumps 3m in one day and falls back 2m in the same day. How many days will it take for the frog to come out of the well?

Question 18. A man counted his animals, 80 heads and 260 legs (ducks and goats). how many goats are there?

Solution:
Let number of ducks be x, and
Number of goats be y
x + y = 80 ---(1), because ducks and goats have only one head each.
2x + 4y = 260 ---(2), because ducks have 2 legs and goats have 4 legs.

(1) x 2 = 2x + 2y = 160
(2) x 1 = 2x + 4y = 260
(1)- (2)
-2y = -100
y = 50 = number of goats.
x = 30 = number of ducks.

Question 19. If an integer "k" is divisible by 2,5 and 13. What is the next number to “k” that is divisible by all the three given numbers?
(a) k+13 (b) k+130 (c) 2k (d) 2k+13

Solution:
If “k” is divisible by 2, 5 and 13, then it must be the multiple of LCM of 2, 5 and 13.
Next number to k which is divisible by 2, 5 and 13 is = k + LCM(2, 5, 13)
LCM(2, 5, 13) = 130
The next number to k is = K+130

Question 20. There are 9 cities numbered 1 to 9. From how many cities the flight can start so as to reach the city 8 either directly or indirectly such that the path formed is divisible by 3?

I n f o s y s

Solution:
Effective distance per day = 1m.
Step 1: Subtract jumping up distance from total height.
30 – 3 = 27
If the frog climbs 1m per day, it will take 27/1 = 27 days for 27 meters.
On the 28th day, it will climb 3m and reach the top.
Total number of days required = 28 days.

Solution:
Trial and error method.
Starting from city 1:
Route 1 to 8; Sum = 9
Starting from city 2:
Route 2 to 5 to 8; Sum = 15
Starting from city 3:
Route 3 to 4 to 8; Sum = 15
Starting from city 4:
Route 4 to 6 to 8; Sum = 18
Starting from city 5:
Route 5 to 2 to 8; Sum = 15
Like this one can form a route with sum equal to multiple of three from any city except starting from city 8.

The number of cities the flight can start
is = 8

Question 21. A family I know has several children. Each boy in this family has as many sisters as brothers but each girl has twice as many brothers as sisters. How many brothers and sisters are there?

Solution:
Assume number of boys = b
Assume number of girls = g
Number of brothers for a boy = b-1
b – 1 = g ---(1)
Number of sisters for a girl = g-1
2(g-1) = b ---(2)
Substitute (2) in (1)
2g – 2 – 1 = g
g = 3
b = 4
Number of brothers = 4
Number of sisters = 3

Question 22. In a class of 150 students 55 speak English; 85 speak Telugu and 30 speak neither English nor Telugu.
i. How many speak both English and Telugu?

ii. How many speak only Telugu?
iii. How many speak at least one of the two languages from English and Telugu?

Solution:
Out of 150, 30 does not speak English or Telugu.
So, 150 – 30 = 120 persons speak English or Telugu or both.
Persons speaking both English and Telugu
= 55 + 85 – 120 = 20
Persons speaking only English
= 55 – 20= 35
Persons speaking only Telugu
= 85 – 20 = 65

Question 23. Find the remainder

**when
the
number 12345678910111213...178179
is divided by 180.**

Solution:
A number is divisible by 180 if the
number is divisible by 4, 5 and 9.
If divided by 4, it gives reminder = 3
If divided by 5, it gives reminder = 4
Sum of the digits = 179 x 180/2 =
16110
Sum of the digits is divisible by 9, so the
reminder when divided by 9 = 0.
The first multiple of 9 which when
divided by 4 and 5 leaves a reminder 3
and 4 respectively is 99.
So
the
reminder
when
12345678…..178179 is divided by 180 is
99.

Question 24. The least perfect square, which is divisible by each of 21, 36 and 66 is:

Solution:
To be divisible by 21, 36 and 66, the
number should be their LCM.
Taking LCM using prime factors:
21 = 31 x 71

36 = 22 x 32

66 = 21 x 31 x 111

LCM = product of
each prime factor
with its highest power
Therefore LCM = 22 x 32 x 71 x 111

= 2 x 2 x 3 x 3 x 7 x 11
To make it as a perfect square we have
to multiply the LCM with 7 x 11
Perfect square value
= 2x2x3x3x7x7x11x11
= 213444

I n f o s y s

Question 25. If the digits of my present age are reversed then I get the age of my son. If 1 year ago my age was twice as that of my son. Find my present age.

Solution:
Let my age be 10x + y

My son’s age is 10y + x
One year ago,
10x + y – 1 = 2(10y + x – 1)
10x -2x + y – 20y – 1 + 2 = 0
8x – 19y = -1
8x = 19y – 1
x = (19y – 1)/8
For y = 3, we get x as an integer
x = (19 x 3 – 1)/8 = 7
My age is 10x7 + 3 = 73
My son’s age is = 37